”I hope they raise their prices”:

Tariff pass-through and markups in the USA

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Motivation

Figure: Tariffs are here,... now who pays the bill?

Tariff Pass-through

Amiti, Redding, and Weinstein (2019) gives an overview on the literature and simple estimates:

- Pass-through to import prices: Complete
- Pass-through to domestic prices: Incomplete
- Loss of variety: Amplifies price impact
- Retaliatory tariffs: Loss of output

Fajgelbaum et al. (2020) provides a coherent framework and finds complete pass-through to import prices. Uses perfect competition, flexible prices and sectoral/ product data.

Firm-level pricing with international shocks

Amiti, Itskhoki, and Konings (2019) find incomplete exchange rate pass-through to domestic prices and show conditions under which this will occur:

- Firm heterogeneity in sensitivity to cost shocks and markup elasticity
- Positive correlation between these two

Baslandze and Fuchs (2025) find that delays in shipping operations affect consumer prices through ...

- inventory depletion and holding costs
- strategic pricing complementarities with competitor firms

Microfoundations and Tariffs

Felbermayr, Jung, and Larch (2015) show that welfare effects on tariffs ...

- depend on whether they are seen as shifters of cost (tax on marginal costs) or demand (tax on export price).
- are stronger under firm heterogeneity.

Grossman, Helpman, and Redding (2024) consider supply chain disruptions through tariffs with firm search and matching.

- With unchanged sourcing (small tariff) and elastic demand for final goods, cost for intermediates may increase more than the tariff.
- With new optimal sourcing (large tariff), relocation to higher-cost options occurs. Welfare losses are smaller with reshoring.

Research Gap

- Evidence points to complete pass-through to import prices, incomplete pass-through to domestic prices
- Firm evidence on domestic price impacts are usually derived under "iceberg" cost shocks, not tariffs
- Empirical evidence with firm adjustment to tariffs is focused on specific sectors - like washing machines - or relies heavily on quantitative models

=> Goal 1: Use firm-level data to assess tariff pass-through to import prices

=> Goal 2: Leverage insights to make statements about domestic prices

Data

- **Panjiva:** Transaction records of maritime shipping
  - Data sourced through customs agencies with information on content, shipper, consignee and carrier
  - Coverage from July 2007 and primarily transactions involving US ports
- **Orbis:** Firm-level balance sheet data
  - Mostly data on large firms
  - Material cost, investment, debt, revenue sources, etc.
- **Tariffs:** Tariff schedules with implementation date
  - Coded at the 10-digit product level
  - Available from US Government or WTO

Empirical Framework

Theory

Set-up as in Fajgelbaum et al. (2020) as nested CES demand system but with firm-level imports as varieties. Results in an unbalanced panel with more FEs.

$$\Delta m_{igt} = \eta^{m}_{it} + \eta^{m}_{i} + \eta^{m}_{gt} + \eta^{m}_{ct} + \eta^{m}_{cs} - \sigma \Delta p_{igt} + \varepsilon^{m}_{igt} \tag{1}$$

$$\Delta p^{*}_{igt} = \eta^{p*}_{it} + \eta^{p*}_{i} + \eta^{p*}_{gt} + \eta^{p*}_{ct} + \eta^{p*}_{cs} - \omega^{*} \Delta m_{igt} + \varepsilon^{p*}_{igt} \tag{2}$$

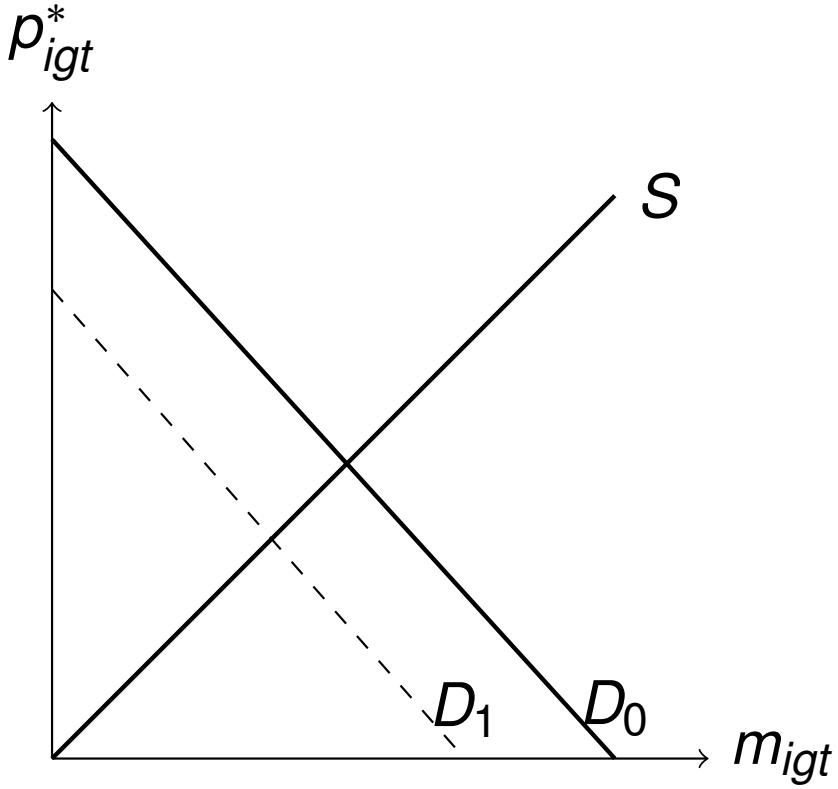
$$p_{igt} = (1 + \tau_{igt}) p^{*}_{igt} \tag{3}$$

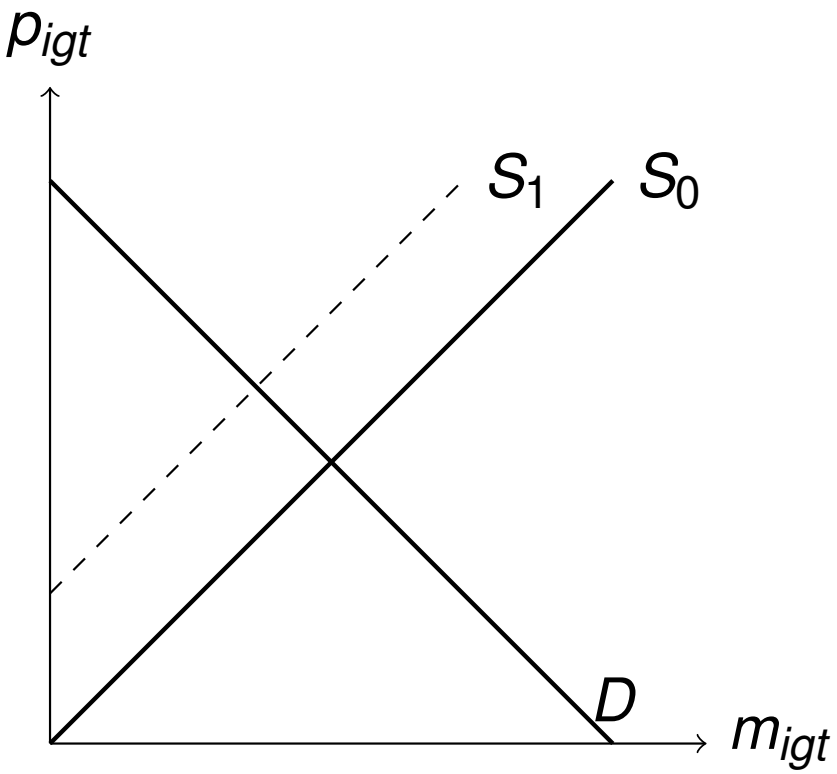
(i importing firm, g product, t time, s sector, c exporting country)

Identification via tariff

Derivation

Identification of elasticities like Zoutman, Gavrilova, and Hopland (2018)





Prices for imports and domestic goods

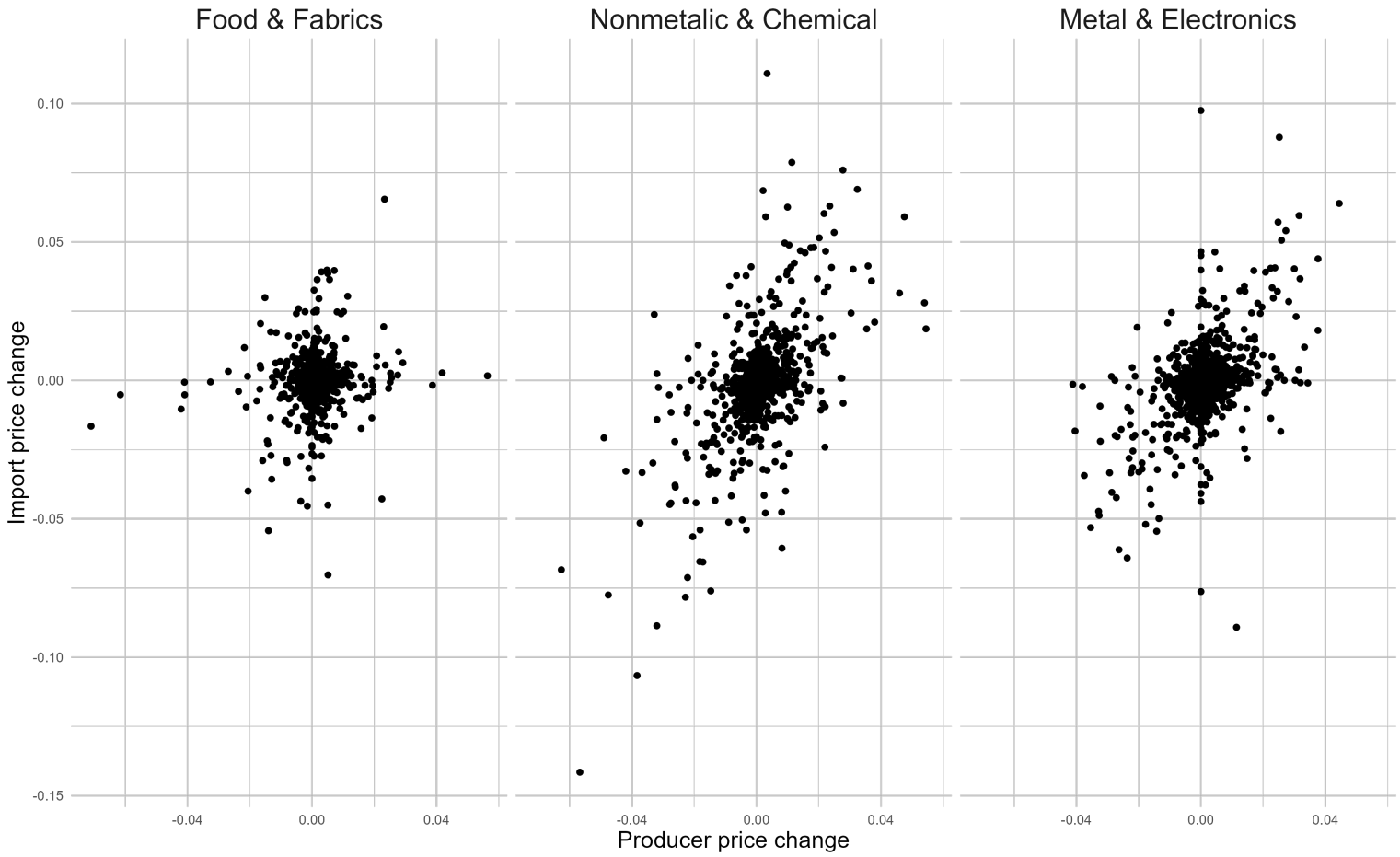


Figure: Scatter plot of price relation for manufacturing sectors

Available Information

- **Prices**: Indices for US import and producer prices
  - Available at product and sector level
  - Some sectors need imputation through aggregated commodity indices
- **FactSet:** Binary indicators for firm-level relationships
  - Collection of public communication of firms regarding relationships to other firms
  - Covers suppliers, customers, competitors, etc.
  - Coverage starting from April 2003 and strongest for large US firms

Domestic price intuition

Pass-through estimation

Empirical frameworks for reduced-form assessment of pass-through exists:

- Very demanding on micro-data, need instruments for marginal costs.
- Usually no price effects of sourcing changes and loss of variety.

Structural frameworks offer general equilibrium assessment:

- Joint modeling of supply chain formation and price propagation becomes non-tractable quickly.
- Imports of intermediates and final goods have widely different modeling approaches.

Conclusion & Open Questions

Old international economics question, but no complete answers. Very relevant, if we do it well.

Step 1: Import prices

- Is this identification via tariff wedge credible at the firm level?
- Does the non-randomness of tariff setting create threats to identification?

Step 2: Domestic prices

- Do we have the relevant firm-level variation to assess tariff mark-up responses?
- Should we prioritise GE approaches over less rigorous identification?

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Nested CES demand

Return

Nested CES demand system with 3 levels as baseline:

$$P_{Ms}M_s = E_s A_{Ms}\left(\frac{P_{Ms}}{P_s}\right)^{1-\kappa} \tag{4}$$

$$p_{Mg}m_g = P_{Ms}M_s a_g\left(\frac{p_{Mg}}{P_{Ms}}\right)^{1-\eta} \tag{5}$$

$$m_{ig} = m_g a_{ig}\left(\frac{p_{ig}}{p_{Mg}}\right)^{-\sigma} \tag{6}$$

- For Fajgelbaum et al. (2020) the lowest nest is imports for product g from country i
- We modify this to imports of firm i in product g from country c.
- Implies each firm importing each product from one country only (hence c in index redundant).

Exclusion restrictions: Standard & Ramsey

Return

Structural demand-supply system, with z_{it} as function of tax rate:

$$y_{it} = \varepsilon^S p_{it} + \eta z_{it} + \Gamma^S x_{it} + \nu_{it}^S \tag{7}$$

$$y_{it} = \varepsilon^D p_{it} + \gamma z_{it} + \Gamma^D x_{it} + \nu_{it}^D \tag{8}$$

Reduced form:

$$\begin{bmatrix} y_{it} \\ p_{it} \end{bmatrix} = \begin{bmatrix} \pi_{zy} \\ \pi_{zy} \end{bmatrix} z_{it} + \begin{bmatrix} \Pi_{xy} \\ \Pi_{xy} \end{bmatrix} x_{it} + \xi_{it}, \qquad \begin{bmatrix} \pi_{zy} \\ \pi_{zy} \end{bmatrix} = \begin{bmatrix} \frac{\varepsilon^S \gamma - \varepsilon^D \eta}{\varepsilon^S - \varepsilon^D} \\ \frac{\gamma - \eta}{\varepsilon^S - \varepsilon^D} \end{bmatrix} \tag{9}$$

- SER: If the tax τ_{it} is levied on the demand side, then $\eta = 0$.
- RER: Demand depends only on the price after taxation. Hence, $z_{it} \equiv log(1 + \tau_{it})$ and $\gamma = \varepsilon^D$.

Marginal costs and mark-ups

Return

Amiti, Itskhoki, and Konings (2019) deliver a flexible baseline for assessing pass-through with strategic complementarities in pricing and firm heterogeneity:

$$p_{it} = mc_{it} + \mathcal{M}_i(p_{it}, p_{-it}; \xi_t) \tag{10}$$

$$dp_{it} = dmc_{it} + \frac{\partial \mathcal{M}_i(p_t; \xi_t)}{\partial p_{it}} dp_{it} + \sum_{j \neq i} \frac{\partial \mathcal{M}_i(p_t; \xi_t)}{\partial p_{jt}} dp_{jt} + \varepsilon_{it} \tag{11}$$

$$dp_{it} = \frac{1}{1 + \Gamma_{it}} dmc_{it} + \frac{\Gamma_{-it}}{1 + \Gamma_{it}} dp_{-it} + \varepsilon_{it} \tag{12}$$

$$dp_{it} = \alpha_{it} dmc_{it} + \gamma_{it} dp_{-it} + \varepsilon_{it} \tag{13}$$